ENSEMBLE LEARNING REFERENCE GUIDE

Random Forest (Bagging): Step-by-Step Worked Solution

A complete, clear trace of Bootstrap Sampling, Feature Subsampling, Decision Rules, OOB Error, and Majority Voting.

1. The Complete Training Dataset (D)

We have $N = 6$ samples and $p = 2$ continuous features: **Age (X_1)** and **Income (X_2)**. The target is **Approved ($y \in \{0, 1\}$)**.

Sample ID	X_1 (Age)	X_2 (Income $k)	True Label y	Interpretation
S1	25	40	0	Younger, lower income → Not Approved
S2	35	60	0	Moderate age and income → Not Approved
S3	45	80	1	Higher age and income → Approved
S4	20	70	0	Young age despite good income → Not Approved
S5	50	30	1	Older age despite lower income → Approved
S6	55	90	1	High age and highest income → Approved

2. Core Concepts: How Random Forest Works (Bagging + Feature Selection)

A. Bootstrap Sampling (Rows): For each tree, draw $N = 6$ samples *with replacement*. Some samples repeat, and the unpicked ones (~36.8%) become the **Out-of-Bag (OOB)** set.

B. Feature Subsampling (Columns): At each split node, pick $m = \lfloor\sqrt{p}\rfloor = \lfloor\sqrt{2}\rfloor = 1$ feature randomly out of $\{X_1, X_2\}$. The tree can only split on that drawn feature.

C. Finding Best Split: Find the threshold t on the selected feature that best separates the in-bag training rows into class 0 and class 1.

3. Building B = 3 Trees Step-by-Step

Tree	Bootstrap Sample (In-Bag Rows)	Leftover Out-of-Bag (OOB)	Random Feature	Learned Decision Rule & Why
Tree 1 (T_1)	Draws: **{S1, S2, S2, S3, S5, S6}** (S2 picked twice; S4 omitted)	**{S4}** (Only S4 was not drawn)	**X_1 (Age)** (Drawn from $\{X_1, X_2\}$)	**If Age > 40 → 1, else 0** Ages in-bag are: 25, 35, 45, 50, 55. A split at 40 perfectly isolates {S1, S2, S2} (all 0) from {S3, S5, S6} (all 1).
Tree 2 (T_2)	Draws: **{S1, S3, S4, S4, S6, S6}** (S4, S6 picked twice; S2, S5 omitted)	**{S2, S5}** (S2 and S5 were not drawn)	**X_2 (Income)** (Drawn from $\{X_1, X_2\}$)	**If Income > 75 → 1, else 0** Incomes in-bag: 40(y=0), 70(y=0), 80(y=1), 90(y=1). Threshold 75 cleanly separates income ≤ 70 (y=0) from > 75 (y=1).
Tree 3 (T_3)	Draws: **{S1, S2, S3, S4, S5, S5}** (S5 picked twice; S6 omitted)	**{S6}** (Only S6 was not drawn)	**X_1 (Age)** (Drawn from $\{X_1, X_2\}$)	**If Age > 40 → 1, else 0** Ages in-bag: 20(y=0), 25(y=0), 35(y=0), 45(y=1), 50(y=1). Threshold 40 perfectly splits ≤ 35 (y=0) and ≥ 45 (y=1).

4. Out-of-Bag (OOB) Evaluation (True Testing Without a Validation Set)

Rule: *Each sample is evaluated ONLY by trees that did NOT include it in their training bootstrap.*

Sample	True y	Trees Where Sample is OOB	Individual OOB Predictions	Majority Vote	Evaluation
S1	0	None (present in T_1, T_2, T_3)	–	–	Not OOB (excluded from evaluation)
S2	0	**T_2 only**	T_2: Income=60 ≤ 75 → 0	**0**	**Correct (0 == 0)**
S3	1	None (present in T_1, T_2, T_3)	–	–	Not OOB (excluded from evaluation)
S4	0	**T_1 only**	T_1: Age=20 ≤ 40 → 0	**0**	**Correct (0 == 0)**
S5	1	**T_2 only**	T_2: Income=30 ≤ 75 → 0	**0**	**Misclassified (Predicted 0, True 1)**
S6	1	**T_3 only**	T_3: Age=55 > 40 → 1	**1**	**Correct (1 == 1)**

OOB Error Calculation: Evaluated samples are {S2, S4, S5, S6} (4 total). 3 out of 4 are correct; 1 is wrong (S5). **OOB Error Rate = 1 / 4 = 25.0%** (Equivalent to **75.0% OOB Accuracy**).

5. Inference: Predicting on a New Unseen Person

A new customer arrives: **Age = 42 (X_1), Income = $72k ($X_2$)**. All trees vote equally:

- **Tree 1 (Rule: Age > 40 → 1):** Age is 42 > 40 → **Votes 1 (Approved)**
- **Tree 2 (Rule: Income > 75 → 1):** Income is 72 ≤ 75 → **Votes 0 (Not Approved)**
- **Tree 3 (Rule: Age > 40 → 1):** Age is 42 > 40 → **Votes 1 (Approved)**

Ensemble Majority Vote = mode{ 1, 0, 1 } = 1 (Approved). With 2 out of 3 votes, the applicant is Approved.